Foundations of ADCS

2. Attitude Determination System

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Foundations of ADCS

https://github.com/risherlock/ADCS

0. Course introduction
1. Preliminaries of ADCS
2. **Attitude determination system**
3. Inorbit models
4. Attitude control system
5. ADCS simulation
6. Discussions and conclusion

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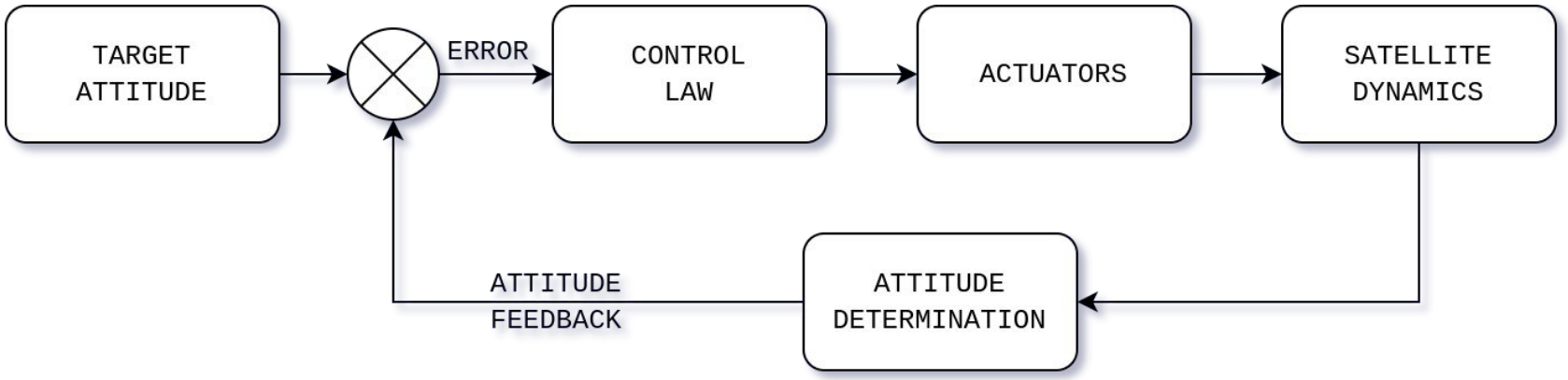
Review: Attitude measurement

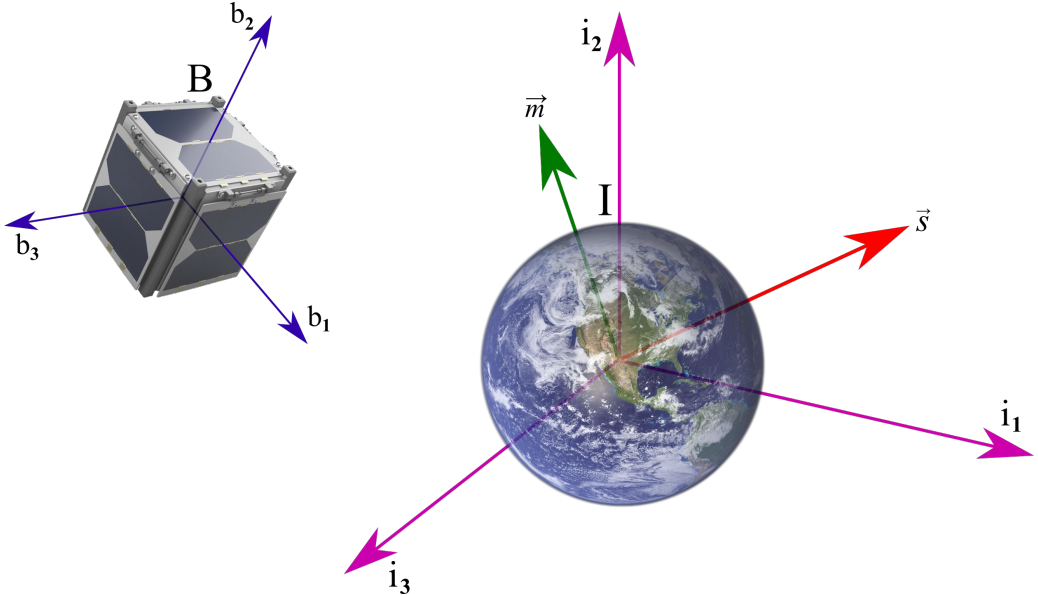
1. Notion of attitude measurement requires at least *two coordinate frames* and a *attitude parameter*
2. Attitude of body frame is expressed relative to the reference frame
3. Simple and straightforward: Euler angles
4. Mathematical beauties: Rotation matrices and quaternions
   a. Consecutive rotation
   b. Coordinate transformation

Our position in the big picture

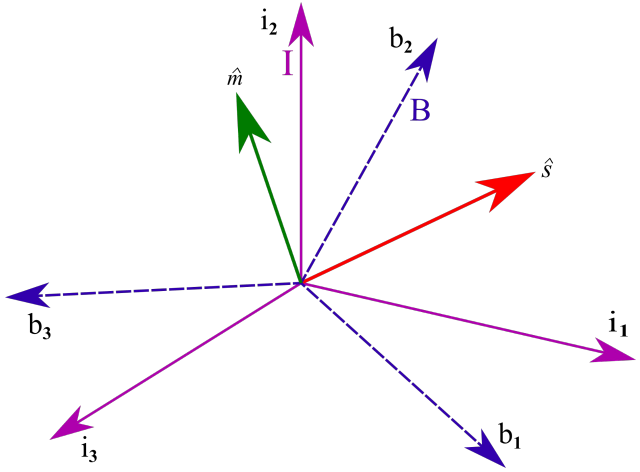


In the last lecture, we quantified attitude. Now we will discuss how one determines feedback attitude.

Vector observations



2-Vector Measurements



$$\hat{s}^B = \boldsymbol{Q}\hat{s}^I$$

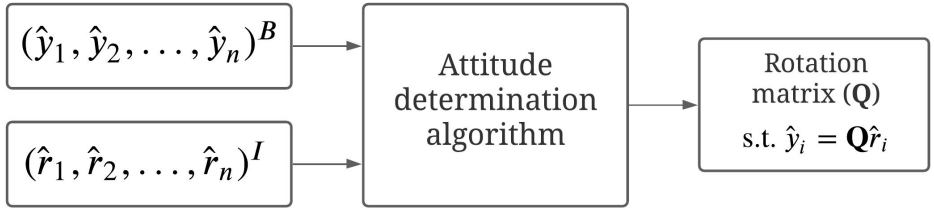
$$\hat{m}^B = \boldsymbol{Q}\hat{m}^I$$

$$\hat{s}, \hat{m} \in \mathbb{R}^3$$

$$\boldsymbol{Q} \in \mathbb{R}^{3\times 3} \text{ and } \boldsymbol{Q}^{-1} = \boldsymbol{Q}^T$$

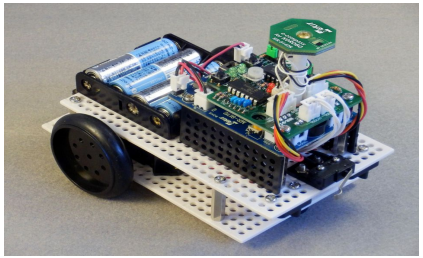
We need at least two vector pairs to solve for rotation matrix.

Attitude Determination

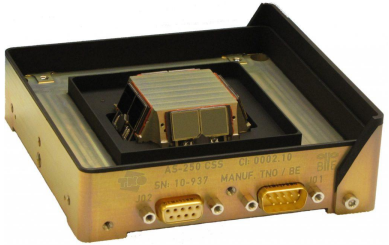


Attitude determination is the process which takes a set of vectors in *body frame* and corresponding vectors in *reference frame* to output attitude parameters.

Vector measuring sensors

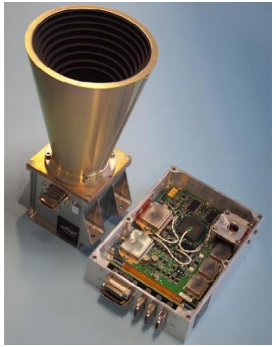


Magnetometer



Sun sensor

Earth sensor



Star sensor

Motivation of TRIAD

Before diving into the TRIAD algorithm itself, let us examine the motivation behind it. Let two sets of orthonormal right-handed unit vectors $\{\hat{\boldsymbol{v}}\} \equiv \{\hat{\boldsymbol{v}}_1,\, \hat{\boldsymbol{v}}_2,\, \hat{\boldsymbol{v}}_3\}$ and $\hat{\boldsymbol{w}} \equiv \{\hat{\boldsymbol{w}}_1,\, \hat{\boldsymbol{w}}_2,\, \hat{\boldsymbol{w}}_3\}$ in the reference frame and body fixed frame respectively. The orthogonal direction cosine matrix $\boldsymbol{Q}$ that projects the vectors $\{\hat{\boldsymbol{v}}\}$ into $\{\hat{\boldsymbol{w}}\}$ is given by,

$$\boldsymbol{Q}_{ij} = \hat{\boldsymbol{w}}_i \,.\, \hat{\boldsymbol{v}}_j$$

$$\boldsymbol{Q} \equiv [\hat{\boldsymbol{w}}_1,\, \hat{\boldsymbol{w}}_2,\, \hat{\boldsymbol{w}}_3][\hat{\boldsymbol{v}}_1,\, \hat{\boldsymbol{v}}_2,\, \hat{\boldsymbol{v}}_3]^{\mathsf{T}} = \sum_{i=1}^{3} \hat{\boldsymbol{w}}_i\, \hat{\boldsymbol{v}}_i^{\mathsf{T}}$$

It can be concluded that using three orthogonal unit vectors in reference frame and their corresponding vectors in the body fixed frame, we can determine the attitude matrix $\boldsymbol{Q}$.

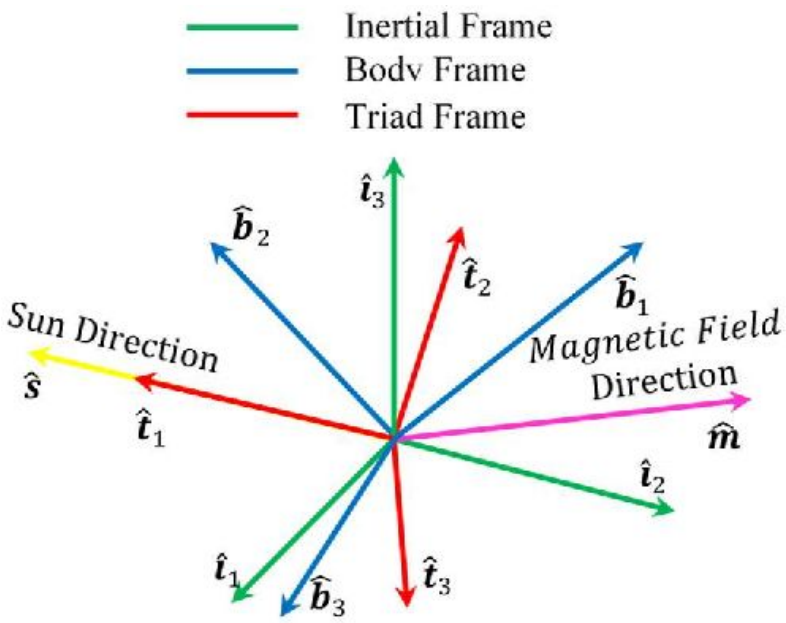
The TRIAD method utilizes this concept to determine attitude matrix using sensor measurements. The algorithm uses the two pairs measurement-reference vector pairs to establish two orthogonal unit vector triads $\{\hat{\boldsymbol{w}}\}$ and $\{\hat{\boldsymbol{v}}\}$ in respective frames. Then these vector triads are used to determine attitude matrix.

TRIAD

$$\hat{\boldsymbol{v}}_1 = \hat{\boldsymbol{r}}_1, \qquad \hat{\boldsymbol{v}}_2 = \frac{\hat{\boldsymbol{r}}_1 \times \hat{\boldsymbol{r}}_2}{||\hat{\boldsymbol{r}}_1 \times \hat{\boldsymbol{r}}_2||}, \qquad \hat{\boldsymbol{v}}_3 = \hat{\boldsymbol{r}}_1 \times \hat{\boldsymbol{v}}_2$$

$$\hat{\boldsymbol{w}}_1 = \hat{\boldsymbol{b}}_1, \qquad \hat{\boldsymbol{w}}_2 = \frac{\hat{\boldsymbol{b}}_1 \times \hat{\boldsymbol{b}}_2}{||\hat{\boldsymbol{b}}_1 \times \hat{\boldsymbol{b}}_2||}, \qquad \hat{\boldsymbol{w}}_3 = \hat{\boldsymbol{b}}_1 \times \hat{\boldsymbol{w}}_2$$

$$\boldsymbol{Q} \equiv [\hat{\boldsymbol{w}}_1,\ \hat{\boldsymbol{w}}_2,\ \hat{\boldsymbol{w}}_3][\hat{\boldsymbol{v}}_1,\ \hat{\boldsymbol{v}}_2,\ \hat{\boldsymbol{v}}_3]^{\mathsf{T}} = \sum_{i=1}^{3} \hat{\boldsymbol{w}}_i\, \hat{\boldsymbol{v}}_i^{\mathsf{T}}$$



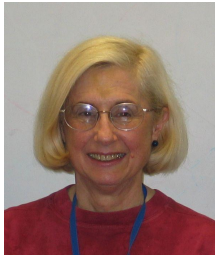
Triad

1. Small Astronomy Satellite (1970), Seasat (1978), Atmospheric Explorer Missions (1878-1979), and Magsat (1979)
2. Simple, computationally efficient, and straightforward to implement
3. Only takes two sensor inputs. Cumbersome for more than two vector pairs
4. Collapses if the measurement vectors are parallel
5. Not an optimal solution

A Least Squares Estimate of Satellite Attitude

Grace Wahba

1965

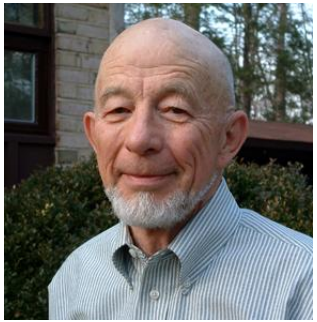


Given two sets of n points $\{\vec{r}_1, \vec{r}_2, ..., \vec{r}_n\}$, and $\{\vec{y}_l, \vec{y}_2, ..., \vec{y}_n\}$, where $n \geq 2$, find the rotation matrix $\boldsymbol{Q}$ (i.e., the orthogonal matrix with determinant $+1$) which brings the first set into the best least squares coincidence with the second. That is, find $\boldsymbol{Q}$ which minimizes

$$\sum_{j=1}^{n} ||\vec{y}_j - \boldsymbol{Q}\vec{r}_j||^2.$$

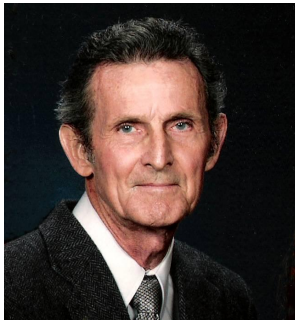
This problem has arisen in the estimation of the attitude of a satellite by using direction cosines $\vec{y}_k$ of objects as observed in a satellite fixed frame of reference and direction cosines $\vec{r}_k$ of the same objects in a known frame of reference. $\boldsymbol{Q}$ is then a least squares estimate of the rotation matrix which carries the known frame of reference into the satellite fixed frame of reference.

Key figures

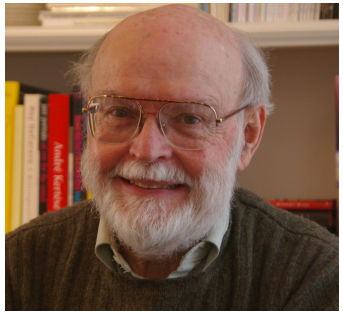


Harold D. Black

Grace Wahba

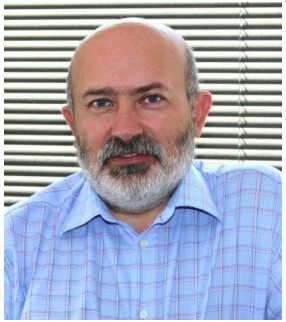


Paul B. Davenport

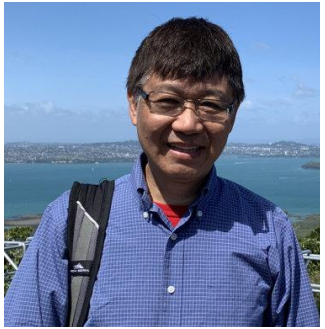


F. Landis Markley

Malcolm D. Shuster



Daniele Mortari



Yaguang Yang

Yang Cheng

Solution of Wahba's problem

$$\boxed{J(\boldsymbol{Q}) = \sum_{i=1}^{n} w_i \, \|\hat{\boldsymbol{b}}_i - \boldsymbol{Q}\,\hat{\boldsymbol{r}}_i\|^2}$$

$$\begin{aligned}
J(\boldsymbol{Q}) &= \frac{1}{2}\sum_{i=1}^{n} w_i \, \|\hat{\boldsymbol{b}}_i - \boldsymbol{Q}\,\hat{\boldsymbol{r}}_i\|^2 \\
&= \frac{1}{2}\sum_{i=1}^{n} w_i \, (\|\hat{\boldsymbol{b}}_i\|^2 - 2\hat{\boldsymbol{b}}_i \,.\, \boldsymbol{Q}\,\hat{\boldsymbol{r}}_i + \|\boldsymbol{Q}\hat{\boldsymbol{r}}_i\|^2) \\
&= \frac{1}{2}\sum_{i=1}^{n} w_i \, (2 - 2\hat{\boldsymbol{b}}_i \,.\, \boldsymbol{Q}\,\hat{\boldsymbol{r}}_i) \\
&= \sum_{i=1}^{n} w_i - \sum_{i=1}^{n} w_i \, \hat{\boldsymbol{b}}_i^{\intercal} \boldsymbol{Q}\,\hat{\boldsymbol{r}}_i \\
&= \lambda_0 - \sum_{i=1}^{n} w_i \, \hat{\boldsymbol{b}}_i^{\intercal} \boldsymbol{Q}\,\hat{\boldsymbol{r}}_i
\end{aligned}$$

Expansion of cost function

$$\begin{aligned}
\sum_{i=1}^{n} w_i \, \hat{\boldsymbol{b}}_i^{\intercal}(\boldsymbol{Q}\,\hat{\boldsymbol{r}}_i) &= \sum_{i=1}^{n} \mathrm{tr}(w_i \, \boldsymbol{Q}\,\hat{\boldsymbol{r}}_i \, \hat{\boldsymbol{b}}_i^{\intercal}) \\
&= \mathrm{tr}(\boldsymbol{Q} \sum_{i=1}^{n} w_i \, \hat{\boldsymbol{r}}_i \, \hat{\boldsymbol{b}}_i^{\intercal}) \\
&= \mathrm{tr}(\boldsymbol{Q}\,\boldsymbol{B}^{\intercal})
\end{aligned}$$

where,

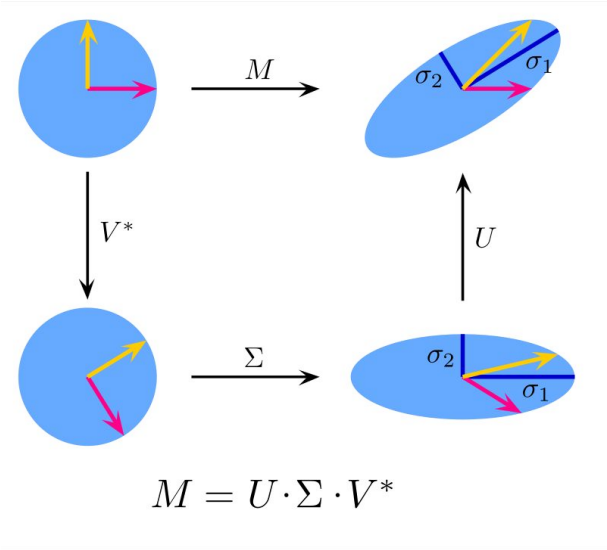
$$\lambda_0 = \sum_{i=1}^{n} w_i \quad \text{and,} \quad \boldsymbol{B} = \sum_{i=1}^{n} w_i \, \hat{\boldsymbol{b}}_i \, \hat{\boldsymbol{r}}_i^{\intercal}$$

Simplification of second term. **B** is called the *attitude profile matrix* and w_i are positive weights.

Attitude profile matrix

$$\boldsymbol{B} = \sum_{i=1}^{n} w_i \, \hat{\boldsymbol{b}}_i \, \hat{\boldsymbol{r}}_i^{\mathsf{T}}$$

SVD



Singular Value Decomposition (SVD) is a mathematical technique that decomposes a given matrix into three separate matrices: a left orthogonal matrix, a diagonal matrix, and a right orthogonal matrix.

Matrix solution

$$\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\intercal = \boldsymbol{B}$$

$$\boldsymbol{Q} = \boldsymbol{U}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & |\boldsymbol{U}||\boldsymbol{V}| \end{bmatrix}\boldsymbol{V}^\intercal$$

Q is the output rotation matrix

Eigenvalues and eigenvectors

$$\boldsymbol{K}\boldsymbol{v} = \lambda \boldsymbol{v}$$

where $\boldsymbol{K} \in \mathbb{R}^{n \times n}$, $\boldsymbol{v} \in \mathbb{R}^n$, and $\lambda \in \mathbb{R}$

Quaternion solution

$$\boldsymbol{Z} = \begin{bmatrix} \boldsymbol{B}_{23} - \boldsymbol{B}_{32} \\ \boldsymbol{B}_{31} - \boldsymbol{B}_{13} \\ \boldsymbol{B}_{12} - \boldsymbol{B}_{21} \end{bmatrix}$$

$$\boldsymbol{K} = \begin{bmatrix} \boldsymbol{B} + \boldsymbol{B}^{\intercal} - \mathbb{I}_{3\times 3}\mathrm{tr}(\boldsymbol{B}) & \boldsymbol{Z} \\ \boldsymbol{Z}^{\intercal} & \mathrm{tr}(\boldsymbol{B}) \end{bmatrix}$$

The output attitude quaternion is the eigenvector of matrix **K** associated with its largest eigenvalue

Numerical solution

1. Initialize a rotation matrix (ground truth)
2. Generate unit vector pairs
3. Implement algorithm
4. Compare the output attitude with ground truth

Hardware implementation

Specifications

Sensors : 3-axis compass and accelerometer

Reference frame : NED (North-East-Down) ??

Algorithm : QUEST by Malcolm Shuster

Vector pairs

$\boldsymbol{m}_b$ = 3-axis compass measurement

$\boldsymbol{a}_b$ = 3-axis accelerometer measurement

$\boldsymbol{m}_r$ = Magnetic field in NED: $[\cos\delta, 0, \sin\delta]^T$

$\boldsymbol{a}_r$ = Accelerometer in NED: $[0, 0, 1]^T$

Attitude estimation: *Kalman Filters*

1. Vector measurements is not possible at times. Eg. sun sensors in eclipse
2. More accurate attitude estimate because it incorporates memory of past observations
3. Makes use of a rotational kinematics model to complement sensor measurements
4. Uses the model and sensor noise parameters as the input to the algorithm

References

1. Malcolm shuster homepage: *https://www.malcolmdshuster.com/*
2. Markley, Crassidis - *Fundamentals of Spacecraft Attitude Determination and Control* (2014)
3. Lecture series : *https://github.com/risherlock/ADCS*
4. Attitude estimation : *https://github.com/risherlock/Attitude-Estimation*
5. Wahba's problem : *https://github.com/risherlock/Wahba*

Thank you!